

GURMUKH KHAROD

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EDUCATION

Simon Fraser University

B.Sc., Computer Science - Software Systems

Burnaby, BC, Canada

May 2022 - May 2027

Douglas College

A.S., Computer Science and Information Systems

New Westminster, BC, Canada

Sep. 2018 - Sep. 2021

EXPERIENCE

Student Software Engineer

BlackBerry QNX - Acoustics Engineering Services

Jan. 2026 - Present

Burnaby, BC, Canada

- Developed **embedded acoustics tooling** in **C/C++** across **6 multi-language audio API modules**, supporting vehicle audio systems with up to **12 speakers**, **6 listening presets**, and **3 frequency bands**.
- Implemented **audio post-processing features** using **FFT-based analysis**, **gain matrices**, and channel configuration across **3 filtering stages** to validate playback behavior on embedded speaker paths.
- Set up and validated an end-to-end **audio measurement pipeline** using **PuTTY**, **Ethernet**, **A2B**, **RME** and **Scarlett** interfaces, **6 microphones**, amplifiers, and **12 speakers** to capture sine-sweep responses and observe acoustic details to support repeatable media-tuning automation.
- Constructed a **voice evaluation pipeline** using **Whisper** and **OpenWakeWord** to evaluate **Google** and **Siri** interactions across **linear and residual echo cancellation**, **beamforming**, **audio filtering**, and **FFT based processing**, reducing errors by over **90 percent** under **extreme echo** and **road noise** while meeting **Google speech recognition performance targets** for both **WER** and **FRR**.

Lead Programmer

FIRST Robotics - FRC Team

May 2017 - Jan. 2020

Vancouver, BC, Canada

- Served as **Lead Programmer** for a **60-member team**, building **autonomous C++ routines** and tuning **encoder feedback** with firmware-hardware integration to meet **FRC specifications** and improve autonomous reliability.
- Collaborated with a **cross-functional team** during a **6-week sprint** to deliver the software of a competition ready robot for **regional** and **world championship** events across **North America**, leading to a **1st Place finish at the Victoria Regional** and the **FIRST Chairman's Award**, while executing **rapid triage**, mentoring junior developers, and supporting outreach.

PROJECTS

Decaf Compiler - C++, Flex, Bison/Yacc, LLVM IR, Linux, Git

May 2025 - Dec. 2025

- Built a **C++ compiler** for a C-like language, implementing lexical and syntactic analysis, **symbol tables**, semantic checks, and **LLVM IR**, achieving **100%** coverage on ~1,200 tests with strict invariants and clear diagnostics.

SolarSense - Python, PyTorch, scikit-learn, FastAPI, React, NumPy, Git

May 2025 - Aug. 2025

- Built an **AI solar geospatial forecasting platform** from NASA satellite data, achieving **96 percent** accuracy.

Gesture Tower - C, C++, OpenCV, Bazel, Node.js, Embedded Linux, Git

Sep. 2024 - Aug. 2025

- Co-developed a real-time multiplayer embedded-vision system on **BeagleY-AI** (Linux), implementing capture and inference pipelines in **C/C++** with **OpenCV** for **live embedded gesture recognition**.
- Bridged diagnostics to a React frontend via **TCP**, **UDP**, **HTTP**, and **WebSockets** through a **Node.js** server, using data structures and **observer/visitor** patterns to deliver a synchronized service.

LEADERSHIP

Hackathons and Mentorship - SFU CSSS and UBC CSSS

Sep. 2023 - Oct. 2025

- **System Hacks 2024 (Winner)**: Built *Chaos Keys* in React with a live REST API; won Best Startup.
- **NW Hacks 2025 (Best UI Finalist)**: Co-built *GROC*, implemented Python APIs and SQL. Best UI finalist.
- **Fall Hacks 2025 (Mentor)**: Mentored teams on Git, REST, and deployment, leading to several awards.
- **StormHacks 2025**: Shipped *UniVerse*, a mobile-first social app built with React Native, GeoJSON, and Socket.IO.

TECHNICAL SKILLS

Languages: C, C++, C#, Java, Python, JavaScript, TypeScript, Haskell, Rust, SQL, NoSQL, HTML5, CSS3

Frameworks and Build: React.js, React Native, Node.js, Express.js, Next.js, FastAPI, Spring Boot, GoogleTest

Data, ML, and CV: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Matplotlib, OpenCV, MediaPipe, LSTM

Embedded and Systems: I²C, SPI, GPIO, ADC, PWM, UART, MCU R5, LLVM IR, Flex, Bison, Yacc

Tooling and Platforms: Git, GitHub, GitLab, CI/CD, APIs, JSON, VMs, Netlify, Vercel, Linux, Windows, macOS